**JINZHUANG DOU, Ph.D.**

Aug 17,2026
The University of Alabama at Birmingham
701 19TH St South | Birmingham, AL 35294
https://jinzhuangdou.github.io/doulab.github.io/
jdou@uab.edu
+1(512) 6969846

**RESEARCH INTERESTS**

Statistical Genomics, Computational Biology, AI in Biomedicine

**EDUCATION**

**Ph.D in Genetics** 08/2012 – 06/2015
Ocean University of China, QingDao, China
**M.S. in Applied Mathematics** 08/2009 – 06/2012
Ocean University of China, QingDao, China
**B.S. in Applied Mathematics** 08/2005 – 06/2009
Ocean University of China, QingDao, China

**RESEARCH EXPERIENCE**

**Assistant Professor** 06/2025 -
The University of Alabama at Birmingham, Birmingham, AL, USA
**Data Scientist** 01/2021 -06/2025
MD Anderson Cancer Center, Houston, TX, USA
**Postdoctoral research fellow** 07/2018 – 01/2021
MD Anderson Cancer Center, Houston, TX, USA
**Postdoctoral research fellow** 07/2015 – 07/2018
Genome Institute of Singapore, A*STAR, Singapore

**AWARDS**

"Wall of Science", MD Anderson Cancer Center (2023)

**MAJOR PUBLICATIONS (* co-first author; ^ corresponding author)**

1. Z Wang, K Chen, **J Dou^ (2026).** Single-Cell Indel Detection Enhances Genetic Ancestry and Cellular Lineage Analysis. *Bioinformatics* (submitted).
2. S, R Luo, Z Wang, J Dou, K Chen, R Chen (**2026**). Somatic variant detection in normal tissues from single cell sequencing data. *Cell Systems* (Accepted in Principle)
3. **J Dou,** Y Tan, KH Kock, J Wang, X Cheng, LM Tan, KY Han, CC Hon, WY Park, JW Shin, H Jin, H Chen, L Ding, S Prabhakar, N Navin. K Chen (**2023**). Monopogen: single nucleotide variant calling from single cell sequencing. *Nature Biotechnology*. 2023 Aug 17:1-0 [Highlight in *Nature Genetics*. https://www.nature.com/articles/s41588-023-01544-2]
4. D Wu*, **J Dou***, X Chai*, C Bellis, A Wilm, C.C. Shih, W.W.J Soon, N Bertin, C.B Lin, C.C Khor, M DeGiorgio, S.S Cheng, L Bao, N Karnani, W Hwang, S Davila, P Tan, A Shabbir, A Moh, E Tan, J.N Foo, L.L Goh, K.P Leong, R.S.Y Foo, C Lam, A Richards, C.Y Cheng, T Aung, T Wong, H Ng, M Ackers Johnson, E Aliwarga, K Kim Ban, D Bertrand, J Chambers, D Hui Chan, C Li Chan, M Chee, M Chee, P Chen, Y Chen, E Chew, W Chew, L Chiam, J Chong, I Chua, S Cook, W Dai, R Dora Joo, C Foo, R Goh, A Hillmer, I Irwan, F Jaufeerally, A Javed, J Jeyakani, J Koh, J Koh, P Krishnaswamy, J Kuan, N Kumari, A Lee, S Lee, S Lee, Y Lee, S Leong, Z Li, P Li, J Liew, O Liew, S Lim, W Lim, C Lim, T Lim, C Lim, S Loh, A Lok, C Chin, S Majithia, S Maurer-Stroh, W Meah, S Mok, N Nagarajan, P Ng, S Ng, Z Ng, J Ng, E Ng, S Ng, S Nusinovici, C Ong, B Pan, V Peder Gnana, S Poh, S Prabhakar, K Prakash, I Quek, C Sabanayagam, W See, Y Sia, X Sim, W Sim, J So, D Soon, E Tai, N Tan, L Tan, H Tan, W Tan, M Tandiono, A Tay, S Thakur, Y Tham, Z Tiang, G Toh, P Tsai, L Veeravalli, C Verma, L Wang, M Wang, W Wong, Z Xie, Yeo, L Zhang, W Zhai, Y Zhao, J Liu, C Wang (**2019**) Large-scale whole-genome sequencing of three diverse Asian populations in Singapore. *Cell* 179(3):736-749 [**Cover story**]

5. Li*, V Mohanty*, **J Dou***, Y Huang*, P P. Banerjee, Q Miao, J Lohr, T Vijaykumar, B Knoechel, R Basar, S Liang, M Kaplan, M Daher, E Liu, L Muniz-Feliciano, T J. Laskowski, D Marin, S Mielke, R E. C, Elizabeth J. Shpall, K Chen, K Rezvani (**2023**). Metabolic competition is an important driver of tumor resistance after CAR NK cell therapy and can be overcome by cytokine engineering. *Science Advance* 9(30), eadd6997.
6. **J Dou**, S Liang, V Mohanty, X Cheng, S Kim, J Choi, Y Li, K Rezvani, R Chen, K Chen (**2022**). Unbiased integration of single cell multi-omics data. *Genome Biology* 23(1), pp.1-25
7. **J Dou**, D Wu, L Ding, K Wang, M Jiang, X Chai, D. Reilly, E Tai, J Liu, X Sim, S Cheng, C Wang (**2020**). Using off-target data from whole-exome sequencing to improve genotyping accuracy, association analysis, and polygenic risk prediction. *Brief in Bioinformatics* bbaa084
8. **J Dou***, B Sun*, X Sim, I Irawan, J D Hughes, D F Reilly, E Tai, J Liu, C Wang (**2017**). Estimation of kinship coefficient in structured and admixed populations using sparse sequencing data. *PLoS Genetics* 13: e1007021s.
9. **J Dou***, H Dou*, C Mu, L Zhang, Y Li, J, T Li, X Hu, S Wang, Z Bao (**2017**). Whole-genome restriction mapping by "sub haploid"-based RAD sequencing: an efficient and flexible approach for physical mapping and genome scaffolding. *Genetics* 206:1237-1250. [**Cover story**]
10. **J Dou**, X Zhao, X Fu, W Jiao, N Wang, L Zhang, X Hu, S Wang, Z Bao (**2012**). Reference-free SNP calling improved accuracy by preventing incorrect calls from repetitive genomic regions. *Biology Direct* 7:17.
11. **J Dou**, X Li, Q Fu, W Jiao, Y Li, T Li, Y Wang, X Hu, S Wang, Z Bao (**2016**). Evaluation of the 2b-RAD method for genomic selection in scallop breeding. *Scientific Report* 6:19244.
12. W Jiao*, X Fu*, **J Dou***, H Li, H Su, J Mao, Q Yu, L Zhang, X Hu, X Huang, Y Wang, S Wang, Z Bao (**2013**). High-resolution linkage and quantitative trait locus mapping aided by genome survey sequencing: building up an integrative genomic framework for a bivalve mollusk. *DNA Research* 21 (1): 85-101. [**Editor's choice**]

## ACTIVE GRANT AWARDS

| | | |
|---|---|---|
| NextGen Scholar Grant (O'Neal Comprehensive Cancer Center; UAB) | $50,000 | 06/2026-06/2028 |
| Start-up Support for New Faculty Recruitment Package (UAB) | - | 06/2025- |

## INVITED TALKS

1. ICIBM 2026 — 08/2026, Buffalo, NY; Genetic information inference from Single cell sequencing
2. UAB DBIDS Power Talk — 09/2025; Computational frameworks integrating single-cell data with GWAS to identify causal signals
3. ICIBM 2025 — 08/2025, Columbus, Ohio; Ancestry-informed single-cell genomics and computational modeling.
4. Texas Single Cell Seminar —10/2023, Zoom; Single nucleotide variant calling from single cell sequencing using Monopogen.
5. Texas Single Cell Seminar —05/2021, Zoom; Unbiased integration of single cell multi-omics data
6. CZI Seed Networks 2020 Annual Meeting—11/2020, Zoom; Unbiased integration of single cell multi-omics data
7. Texas Single Cell Seminar —02/2020, Houston, TX; Single-cell characterization of CAR-NK cell exhaustion in transplantation immunotherapy.

## SOFTWARES

1. *CCAdge* (CCA-based DEGs analysis in single cell data). Identify biologically varied genes in population level single cell data.
   https://github.com/KChen-lab/CCAdge
2. *Monopogen* (Single nucleotide variant calling from single cell sequencing). Achieve SNV calling from single cell RNA sequencing, single cell ATAC sequencing.
   https://github.com/KChen-lab/Monopogen
3. *bindSC* (Bi-order integration of single cell multi-omics data). Achieve bi-order integration (in silico multi-omics data) of single cell RNA sequencing, single cell ATAC sequencing, spatial transcriptomics and CyTOF data.
   https://github.com/KChen-lab/bindSC
4. *WEScall* (A genotype calling pipeline for whole exome sequencing (WES) data). Analyze both target and off-target data to facilitate WES studies with decent amount of off-target data.
   https://github.com/dwuab/WEScall
5. *SEEKIN* (Sequencing-based estimation of kinship and inbreeding coefficients). Estimates the kinship and inbreeding coefficients based on extremely low-sequencing coverage datasets (typically lower than 0.5X).

https://github.com/chaolongwang/SEEKIN

6. *AMMO* (An integrated package for whole-genome restriction mapping and genome scaffolding). Generate 2b-RAD-based restriction maps for scaffolding draft genome assemblies produced by using short Illumina reads or long PacBio reads.
   https://github.com/jinzhuangdou/AMMO
7. *RADtyping* (De novo RAD Genotyping in Mapping Populations). Accurate De Novo codominant and dominant RAD genotyping in mapping populations.
   https://github.com/jinzhuangdou/RADtyping

## TEACHING (^ guest lecture)

1. 01/2026-05/2026 INFO604/704: Next-Generation Sequencing Data Analysis
2. 08/2022 Multiomics integration and feature selection ^, Gulf Coast Consortia (GCC) Single Cell Omics Workshop
3. 11/2025 INFO601/701: Introduction to Bioinformatics^

## TRAINNING & MENTORSHIP

*Graduate students*

Ruofan Mao, MS in computer science from UAB, 08/2025-

*Postdoctoral Fellows*

Xin Liang, Ph.D. in bioinformatics from Harbin Medical University, 09/2025-

## PEER REVIEW EXPERIENCE

**Ad Hoc journal reviewer**

Cell, Nature Biotechnology, Genome Biology, Genome Research, Nucleic Acids Research, Bioinformatics, Biology Direct, PLoS ONE

## INSTITUTIONAL SERVICE

Co-Chair, The Annual Translational and Transformative Informatics Symposium (ATTIS), UAB, 05/2026